

True / False (20 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'A packet filter firewall filters at the' is 'Data link layer'.
2. True or False: The correct answer to '_____ is also a part of darknet that is employed for transferring files anonymously.' is 'Freenet'.
3. True or False: The correct answer to 'Which of the following is an authentication method?' is 'All of the above'.
4. True or False: The correct answer to 'A _____ is a sequential segment of the memory location that is allocated for containing some data such as a character string or an...' is 'buffer'.
5. True or False: The correct answer to 'The _____ is anything which your search engine cannot search.' is 'Haunted web'.
6. True or False: The correct answer to 'Why apps developed in languages like C, C++ is prone to Buffer-overflow?' is 'No string boundary checks in predefined functions'.
7. True or False: The correct answer to 'How does a buffer overflow on the stack facilitate running attacker-injected code?' is 'By writing directly to %eax the address of the code'.
8. True or False: The correct answer to 'A/an _____ is a program that steals your logins & passwords for instant messaging applications.' is 'Trojan-Downloader'.
9. True or False: The correct answer to 'A _____ is a method in which a computer security mechanism is bypassed untraceable for accessing the computer or its information.' is 'backdoor'.
10. True or False: The correct answer to 'Let $I = (S, V)$ be a MAC. Suppose $S(k, m)$ is always 5 bits long. Can this MAC be secure?' is 'It depends on the details of the MAC'.

Long Form Response (20 marks)

Provide thorough reasoning and reference key steps.

1. Write a function to check whether the given number is undulating or not.
2. Write a function to sort a tuple by its float element.

End of Paper